SACHIN ASHOK

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RESEARCH INTERESTS

Systems & Networking, Intelligent Systems, Programmable Networks

EDUCATION

National Institute of Technology, Trichy, India *July 2014 - May 2018*
Bachelor of Technology in **Computer Science and Technology** *CGPA: 8.06*

KV Pangode, Trivandrum, India *Graduated in 2014*
Higher Secondary Education, Grade 12 *96.2%*
Secondary Education, Grade 10 *CGPA: 10.0*

RESEARCH EXPERIENCE

Microsoft Research, India *July 2018 - Present*

Research Fellow, Mobility, Networks and Systems Group
Mentors: Dr. Venkat Padmanabhan, Dr. Johannes Gehrke, Dr. Nagarajan Natarajan

- **iBox: Internet In A Box** (Ongoing)
  - Leading the effort to create a data-informed network simulator which uses training data in the form of I/O packet traces from a target network (e.g. the Internet). The research goal is to use the model to generate "realistic" delay and packet loss output to evaluate new algorithms inexpensively before flighting.
  - Ongoing work on enabling A/B testing with iBox is able to replicate the performance of in-house Skype algorithms in simulation topologies used within Microsoft's test-and-release framework. In particular, iBox trained with data from a specific congestion control variant can match several KPI metrics for unseen variants when run on iBox, within a 10% error margin.
  - iBox modules have been integrated into the NetEm emulator and Skype's in-house simulator in prototype form for Microsoft developers to use.
  - Invited for presentation at **Microsoft TechFest, 2020** (an internal conference where Microsoft Research labs showcase promising work).
- **Hole Based Congestion Control** (Ongoing)
  - Building a congestion control scheme for audio video services which aims to improve throughput while respecting real-time constraints like delay and jitter.
  - In simulation, the congestion control scheme produces 10-20% improvement on throughput while keeping intact the one way delay and packet loss rate faced relative to Skype's existing algorithms.
- **R3Net: Reinforcement Learning for bandwidth control**
  - Part of an effort across MSR India, MSR Asia & Microsoft Redmond to use reinforcement learning for bandwidth estimation & congestion control. Developed network environments to train RL agents.
  - Published at **MLForSystems, NeurIPS (2019).**

Samsung Research, India *May 2017 - July 2017*

Research Intern, Multimedia Group
Mentors: Dr. Pratibha Moogi, Mr. Karthikeyan Somanathan, Mr. Adarshdeep Singh

- **Food localizer**
  - Developed an end to end system to leverage deep learning models using custom convolutional neural networks for localization of food in a given image. Entire system pipeline (image capture to model inference) was pushed to production for Samsung mobile devices.

PUBLICATIONS

Joyce Fang, Martin Ellis, Bin Li, Siyao Liu, Yasaman Hosseinkashi, Michael Revow, Albert Sadovnikov, Ziyuan Liu, Peng Cheng, **Sachin Ashok**, David Zhao, Ross Cutler, Yan Lu, and Johannes Gehrke, "Reinforcement Learning for bandwidth estimation and congestion control in real-time communications", *MLForSystems, NeurIPS 2019*

INDUSTRY EXPERIENCE

Mozilla Organisation *Oct 2017 - March 2017*

Intern, Mozilla Winter of Security
Mentors: Dr. Franziskus Keifer, Mr. Tim Taubert

- **Blake2 support for the Network Security Services library**
  - Implemented a Blake2 (a cryptographic hash function) module for Mozilla's NSS library. Library is used by Google, RedHat, AOL and Mozilla products like Thunderbird.
  - **Among 45 university students chosen worldwide** for internship opportunity.

AWARDS & ACHIEVEMENTS

- **Among 77 students chosen in India by Indian Institute of Science** (IISc) for the 4th Undergraduate Summer School (co-sponsored by Google & Microsoft).
- Selected as an exchange student for research programme by **The Institute of Mathematical Sciences (IMSc) (26 students selected nation-wide)**. Gave a talk on online algorithms to a panel of computer science researchers.
- **Finalist** for Microsoft Build The Shield, a nation-wide contest organized by Microsoft.
- **Finalist** in Pragyan CTF, an international level security contest held by NIT Trichy.
- Awarded **1st prize** in Vortex CTF, a popular capture the flag contest held annually.
- Awarded **1st prize (National)** for Parliamentary Debate (2012).

SELECTED PROJECTS

Code Character [Github]

Arena for AI bot fighting / C++

- Developed the system architecture for a tech contest where contestants write C++ code through various APIs exposed to them to control troops and fight the computer AI in a real time strategy game (like age of empires).
- Designed an efficient communication scheme to enable real time transfer of serialized user actions to the back-end handling the main game state to prevent lag during gaming (reduced lag by 13x).
- Developed the computer AI using state-driven AI design for the game simulator (a multi-threaded C++ engine).
- Event received registrations from over 74 countries and handled more than 1000 concurrent users at peak load.

Google-Recaptcha Breaker [Github]

System to programmatically break Google-Recaptcha / Python

- Created an automated, multi-threaded tool that uses cookie manipulation, reverse image search and object detection to solve the image captcha.

Auto-Solver [Github]

AI for solving Yinsh, a board game / C++

- Developed a game AI using Negascout and Monte Carlo Tree Search to conclusively beat human participants by a margin of 3-0.
- Designed a game simulator using bit boards (a specialized bit-array data structure) for efficient state space exploration in tree-based heuristic search. The game tree searching was sped up by 1.5x.

SKILLS & LANGUAGES

Languages	C++, C, Python
Software	NS2/3, MongoDB, Protobuf, gRPC, CMake

OTHER EXPERIENCE

- Gave a talk on "Making an Interpreter" to university students (Level: Sophomore, Junior).
- Conducted python workshop for university students (Level: Freshman, Sophomore, Junior).